

Matrix functions and stability

Degree coverage with seven matrix multiplications

Difficulty: challenging **Importance:** interesting to the community

Problem statement

Start with the scalar polynomials $1, x \in \mathbb{C}[x]$. Allow arbitrarily many complex linear combinations of already computed polynomials and at most seven multiplications of two already computed polynomials. Let $\mathcal{P}_7 \subseteq \mathbb{C}[x]_{\leq 128}$ be the set of possible outputs.

Identify a polynomial with its coefficient vector in $\mathbb{C}^{129}$ and let $\overline{\mathcal{P}_7}^Z$ be its Zariski closure: the common zero set of all polynomial equations in the coefficients that vanish throughout $\mathcal{P}_7$. Is

$$\max\{d \in \{0, \dots, 128\} : \mathbb{C}[x]_{\leq d} \subseteq \overline{\mathcal{P}_7}^Z\} = 42?$$

The target concerns closure and the complex coefficient field. Exact representability of every polynomial is a stronger assertion.

Why it matters

For a dense input matrix, matrix products dominate the cost of polynomial evaluation. The conjecture determines how much polynomial degree a budget of seven products can cover, providing a theoretical benchmark for matrix-function evaluation schemes.

References

1. E. Jarlebring and G. Lorentzon, *The polynomial set associated with a fixed number of matrix-matrix multiplications*, arXiv:2504.01500v3 (13 August 2025), §2.2 (closure convention), §5.4, Conjecture 13. [Primary text](#).
2. J. Sastre, J. Ibáñez, J. M. Alonso, and E. Defez, *Beyond Paterson–Stockmeyer: Advancing matrix polynomial computation*, WSEAS Transactions on Mathematics 24 (2025), 684–693, §4 (a five-product construction). [Primary paper](#), [DOI](#).
3. J. M. Alonso, J. Sastre, J. Ibáñez, and E. Defez, *A systematic framework for stable and cost-efficient matrix polynomial evaluation*, arXiv:2603.23143v1 (24 March 2026), abstract and conclusions. [Primary text](#).

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The latest arXiv source remains v3 and labels the displayed equality a conjecture supported by computations. The later papers give constructions and stability procedures without establishing the seven-product degree-coverage claim. Searches included the exact title with 2026, *Jarlebring Lorentzon conjecture seven multiplications 42*, and *Sastre polynomial evaluation 2026*. No resolution was located. A [3 September 2026 author talk](#) also describes higher-cost degree coverage as conjectural. Six-product real/complex formulations have a separate source ambiguity and are deliberately not included here.